Topic 12 — Acid Rain (अम्ल वर्षा)

Covers syllabus (click to expand)

Acid Rain | Causes | Effects | Prevention

Sources baked in: NCERT Geography/Biology, CPCB norms, MoEFCC, CLRTAP/Gothenburg Protocol, Taj Trapezium case, UPPCS PYQs 2018–2025 **Weight:** ★★ Medium — pH traps, CO_2 vs SO_2/NOx, Taj Mahal, treaty confusion **Last verified:** August 2026

Current Affairs (this topic)

- Acid-rain CA is rare; keep SO_2 / NO_x chemistry and dry vs wet deposition stable.
- Do not confuse acid rain with ozone hole (ओजोन छिद्र) or greenhouse trapping.

12.1 Acid Rain

Identity: pH < 5.6 | H_2SO_4 + HNO_3 | Wet + dry deposition | Secondary pollutant

- **Acid rain** is precipitation or atmospheric deposition with **pH below 5.6**, mainly from **sulphuric acid (H_2SO_4)** and **nitric acid (HNO_3)** (नाइट्रिक अम्ल).
- **Normal unpolluted rain** has pH about **5.6**, not 7, because atmospheric **CO_2** forms weak **carbonic acid (H_2CO_3)**.
- Anthropogenic acid rain often reaches **pH 4.0–4.5** in heavily industrial downwind areas.
- The **pH scale** runs 0–14. **pH 7 is neutral**. **Below 7 is acidic**. **Above 7 is basic/alkaline**.
- Each one-unit drop in pH means **10 times more hydrogen ions**, so pH 4 is 100× more acidic than pH 6.
- **Wet deposition** carries acids in rain, snow, fog, or dew.
- **Dry deposition** means **SO_2**, **NO_2**, and sulphate/nitrate particles settle on soil, leaves, and buildings **without rainfall**.
- **SO_2** and **NOx** are **primary pollutants**. Strong acids form in the atmosphere, so acid rain is a **secondary pollutant** phenomenon.
- Acidifying pollutants can travel **hundreds to thousands of km**, causing **transboundary damage**.
- Soils and lakes with **limestone or calcium carbonate** have **buffering capacity** and resist acidification better than granite or thin acidic soils.

PYQ — UPPCS Prelims 2025, Q103

1. Acidic solutions have pH more than 7.
2. Basic solutions have pH less than 7.

A. Only 2

B. Neither

C. Both

D. Only 1

Show answer

Ans: B — Acidic = **< 7**; Basic = **> 7**.

PYQ — UPPCS Prelims 2024, Q108

(A) CO_2, SO_2, and NO_2 dissolve in rainwater and produce acid rain. (R) Air is polluted due to high concentration of CO_2, SO_2, and oxides of nitrogen.

A. Both true; R explains A

B. A false, R true

C. Both true; R not explanation

D. A true, R false

Show answer

Ans: C — All three dissolve (A true). Air pollution statement (R) is true but does **not explain** why **CO_2 gives only weak acid** while **SO_2/NOx give strong acids**.

pH	Meaning
< 5.6	Anthropogenic acid rain threshold
~5.6	Normal rain (CO_2 equilibrium)
7	Neutral distilled water — not natural rain
> 7	Basic/alkaline

12.2 Causes

Identity: SO_2 → H_2SO_4 | NOx → HNO_3 | Coal TPPs + vehicles | CO_2 weak only

- **Sulphur dioxide (SO_2)** is the main source of **sulphuric acid rain**.
- SO_2 comes from burning **sulphur-containing coal and oil** in **thermal power plants, refineries, smelters, brick kilns, and diesel generators**.
- In the atmosphere, SO_2 oxidises and reacts with water to form **H_2SO_4**.
- **Nitrogen oxides (NOx)** form at **high combustion temperatures** when nitrogen in air reacts with oxygen.
- NOx mainly comes from **vehicles, power plants, industrial boilers, and lightning**.
- NOx oxidises to form **HNO_3 (nitric acid)** in the atmosphere.

- **CO_2** dissolves as weak carbonic acid and sets the **pH 5.6 baseline**. It is **not** the main driver of severe acid rain.
- **Coal thermal power plants** are India's largest **SO_2** point sources.
- **Vehicles** are a major **NOx** source in cities.
- **SO_2 is heavier than air** and tends to accumulate near ground-level emission sources.
- Natural sources include **volcanic SO_2** and **lightning NOx**, but anthropogenic sources dominate in industrial regions.
- The **Singrauli–Sonbhadra belt** on the UP–MP border is a major Indian **SO_2 hotspot** from super thermal power plants.

PYQ — UPPCS Prelims 2025, Q43

Which gas/gases is/are lighter than air?

1. Hydrogen chloride
2. Sulfur dioxide
3. Helium

A. 1 and 2

B. Only 3

C. 2 and 3

D. Only 1

Show answer

Ans: B — HCl and SO_2 are **heavier** than air.

Pollutant	Main acid	Main sources
SO_2	H_2SO_4	Coal TPPs, smelters, refineries
NOx	HNO_3	Vehicles, high-temp combustion
CO_2	H_2CO_3 (weak)	Natural baseline only

12.3 Effects

Identity: Aquatic kill | Soil Al^{3+} | Conifers | Taj gypsum | Lichens

- **Aquatic ecosystems** fail when lake/river pH drops **below about 5.0** — fish eggs do not hatch and adult fish die.
- **Aluminium (Al^{3+})** leached from acidified soil is **toxic to fish gills** and plant roots.
- **Soil acidification** leaches essential **calcium (Ca^{2+})** and **magnesium (Mg^{2+})** and releases toxic **Al^{3+}**.
- **Coniferous forests** (pine, spruce, fir) are **most sensitive** to acid rain and acid fog.
- **Crops** suffer from direct leaf damage and nutrient loss in soil.

- **Marble and limestone ($CaCO_3$)** react with sulphuric acid to form **gypsum ($CaSO_4$)**, causing yellow crust and pitting.
- The **Taj Mahal (Agra, UP)** is the classic Indian example — **SO_2 from Mathura refinery and Firozabad industries** damaged white marble.
- **Metal structures** rust faster under acid deposition.
- **SO_2 and NOx** cause **respiratory illness** — asthma, bronchitis, and lung irritation from acid aerosols and $PM_{2.5}$.
- **Lichens** are **bio-indicators of clean air**. They **die in high SO_2** and do **not** thrive in polluted areas.
- Transboundary teaching: **Norway** (and Canada/Scandinavia) are classic heavily affected countries; acidified lakes are sometimes called **"lake killers."**
- The **Acid Rain Information Centre (ARIC)** is associated with **Manchester, England**.
- The **1985 Helsinki Protocol** targeted about a **30%** cut in sulphur emissions / transboundary fluxes.

> **Logic:** — lichens as pollution indicators = true; grow well in polluted areas = **false**.

PYQ — UPPCS Prelims 2025, Q61

1. Lichens are very good indicators of pollution.
2. Lichens grow well in polluted areas.

A. Only 2

B. Neither

C. Both

D. Only 1

Show answer

Ans: D — Statement 1 only.

Receptor	Key impact
Lakes/rivers	Fish kills below pH ~5
Soil	Nutrient leaching, Al^{3+} toxicity
Forests	Conifer needle damage
Marble monuments	Gypsum crust (Taj Mahal)
Lichens	Die in SO_2 pollution

12.4 Prevention

Identity: FGD | Catalytic converters | BS-VI | Gothenburg | Taj Trapezium

- **Flue Gas Desulphurisation (FGD) scrubbers** pass chimney gas through **lime or limestone slurry** to absorb **SO_2** and can remove **90%+** of it.
- **Coal washing** and **low-sulphur coal** cut sulphur before combustion.
- **Catalytic converters** in vehicles convert **NO to N_2** and reduce **NOx** that forms nitric acid.
- **BS-VI norms (2020)** brought **10 ppm sulphur** in petrol and diesel, cutting both SO_2 and NOx precursors.
- **CNG, LPG, and natural gas** produce **negligible SO_2** and lower NOx than coal.
- **CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड) emission standards** under the Air Act cap SO_2 and NOx from TPPs, cement, steel, and refineries.
- **Tall chimneys alone are not prevention** — they only disperse pollution farther.
- The **Gothenburg Protocol (1999)** under **CLRTAP (1979)** sets binding cuts in SO_2 and NOx for transboundary acid rain control in Europe.
- **Montreal Protocol controls ozone**, not acid rain — do not confuse treaties.
- The **Taj Trapezium Zone (~10,400 km^2)** was created to protect the Taj Mahal. The **1996 Supreme Court order** directed polluting industries near Agra to switch to gas or relocate.
- **NCAP (2019)** targets PM reduction in 122 cities with co-benefits for SO_2 and NOx cuts.

Measure	Target
FGD scrubbers	SO_2 from TPP chimneys
Catalytic converters	NOx from vehicles
BS-VI fuel	SO_2 + NOx precursors
CNG/LPG switch	Near-zero SO_2
Gothenburg Protocol	Transboundary SO_2/NOx
Taj Trapezium	Heritage protection near Agra